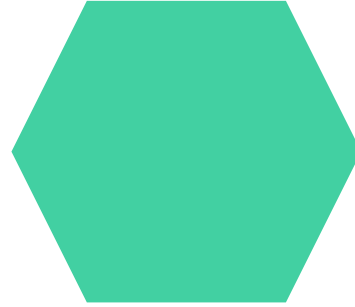




Project Title - Car Market Trends Analysis with Car Dekho Data

AICTE STU ID : STU683a1310cc13e1748636432

[Yashveer Singh]



PROBLEM STATEMENT

- The automobile market contains a large amount of information about cars, including their selling prices, manufacturing details, fuel types, transmission types, and other attributes.
- The objective of this project is to analyze Car Dekho data to identify important patterns and trends in the used-car market.
- The analysis focuses on understanding factors that influence car prices and identifying useful insights from the available dataset.

Main objectives :

- Analyze the Car Dekho dataset.
- Understand car price patterns.
- Explore different car attributes.
- Identify trends and relationships in the data.
- Generate meaningful insights using data analysis and visualization.



Project Description

- This project performs exploratory data analysis (EDA) on Car Dekho data to understand trends in the used-car market.
- The dataset is first explored and prepared for analysis. Different attributes of cars are analyzed to identify patterns related to selling price, present price, fuel type, transmission, ownership, kilometers driven and other available features.
- Python-based data analysis techniques are used to clean, explore and visualize the dataset. The results help in understanding the factors associated with car prices and overall market trends.

Key Analysis Areas :

- Dataset exploration
- Data cleaning and preprocessing
- Descriptive statistics
- Feature-wise analysis
- Price analysis
- Data visualization
- Insights and observations

WHO ARE THE END USERS?

-  Car Buyers - To understand price trends and available options.
-  Car Dealers/Sellers - To understand market patterns and pricing.
-  Automobile Analysts - To analyze used-car market trends.
-  Business Teams - To support data-driven pricing and business decisions.
-  Students & Data Analysts - For learning and studying real-world data analysis.

Technology Used

Programming Language :

- Python

Libraries :

- Pandas
- NumPy
- Matplotlib

Development Environment

- Google Colab

Concepts :

- Exploratory Data Analysis (EDA)
- Statistical Analysis
- Data Visualization



RESULTS

The analysis of the Car Dekho dataset helped identify various patterns and relationships in the used-car market.

Key findings:

- Different factors show variation in car selling prices.
- Car age and usage have an impact on selling price.
- Fuel type and transmission can be compared based on their distribution.
- The dataset provides useful information for understanding used-car market trends.
- Data visualization makes the identified patterns easier to interpret.

[Demo Link](#)

🔍 Commands + Code ▾ + Text | ▶ Run all ▾ ☁



[2]

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```
#Data Analysis Case Study Car Dekho
#In this case study car data is provided as input in the form of csv file
#You have to perform data analysis on it and
#you have to find the answers of questions mentioned in the case study.
#Importing essential packages
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[4]

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```
#Importing the data and taking a look at it
data = pd.read_csv("/Car Market Trends Analysis with Car Dekho Data.csv")
data.head()
```



	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0



[5]
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 `data.tail()`

...	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner	
296	city	2016	9.50	11.6	33988	Diesel	Dealer	Manual	0	
297	brio	2015	4.00	5.9	60000	Petrol	Dealer	Manual	0	
298	city	2009	3.35	11.0	87934	Petrol	Dealer	Manual	0	
299	city	2017	11.50	12.5	9000	Diesel	Dealer	Manual	0	
300	brio	2016	5.30	5.9	5464	Petrol	Dealer	Manual	0	

[6]
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 `#To check the statistical features of the data`
`data.describe() # The statistical features are only for numeric variables`

...	Year	Selling_Price	Present_Price	Kms_Driven	Owner	
count	301.000000	301.000000	301.000000	301.000000	301.000000	
mean	2013.627907	4.661296	7.628472	36947.205980	0.043189	
std	2.891554	5.082812	8.644115	38886.883882	0.247915	
min	2003.000000	0.100000	0.320000	500.000000	0.000000	
25%	2012.000000	0.900000	1.200000	15000.000000	0.000000	
50%	2014.000000	3.600000	6.400000	32000.000000	0.000000	
75%	2016.000000	6.000000	9.900000	48767.000000	0.000000	
max	2018.000000	35.000000	92.600000	500000.000000	3.000000	

From above analysis one can say that -

1. The data here contains cars with manufacturing year 2003 to 2018.
2. The lowest price on which car was sold is INR 10,000/-
3. The highest price on which the car was sold is INR 35 lacs.
4. There are 301 records in the data.


```
#To Look at the variables in detail
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 301 entries, 0 to 300
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Car_Name        301 non-null    object
1   Year            301 non-null    int64
2   Selling_Price   301 non-null    float64
3   Present_Price   301 non-null    float64
4   Kms_Driven       301 non-null    int64
5   Fuel_Type       301 non-null    object
6   Seller_Type     301 non-null    object
7   Transmission    301 non-null    object
8   Owner           301 non-null    int64
dtypes: float64(2), int64(3), object(4)
memory usage: 21.3+ KB
```

```
data.isnull().sum()
```

```
0
Car_Name  0
Year      0
Selling_Price  0
Present_Price  0
Kms_Driven  0
Fuel_Type  0
Seller_Type  0
Transmission  0
Owner       0
dtype: int64
```

From above analysis one can say that -

1. None of the variables or columns here have null values.
2. Variables "Car Name", "Fuel Type", "Seller Type" and "Transmission" non numeric in nature.



[10]

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```
#Checking whether nom numeric variables are categorical in nature
print(data.Car_Name.value_counts())
```

▼

```
*** Car_Name
city          26
corolla altis 16
verna         14
fortuner      11
brio          10
..
Honda Activa 125 1
Hero Hunk        1
Hero Ignitor Disc 1
Hero CBZ Xtreme  1
Bajaj ct 100     1
Name: count, Length: 98, dtype: int64
```

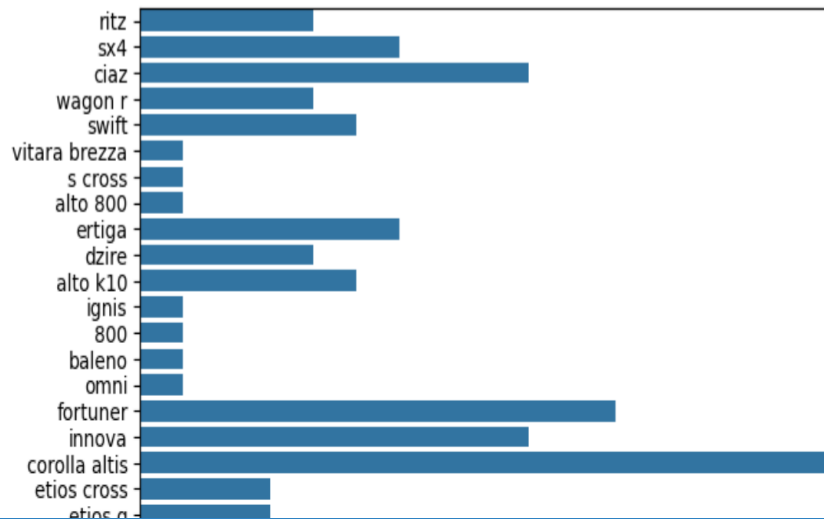
[11]

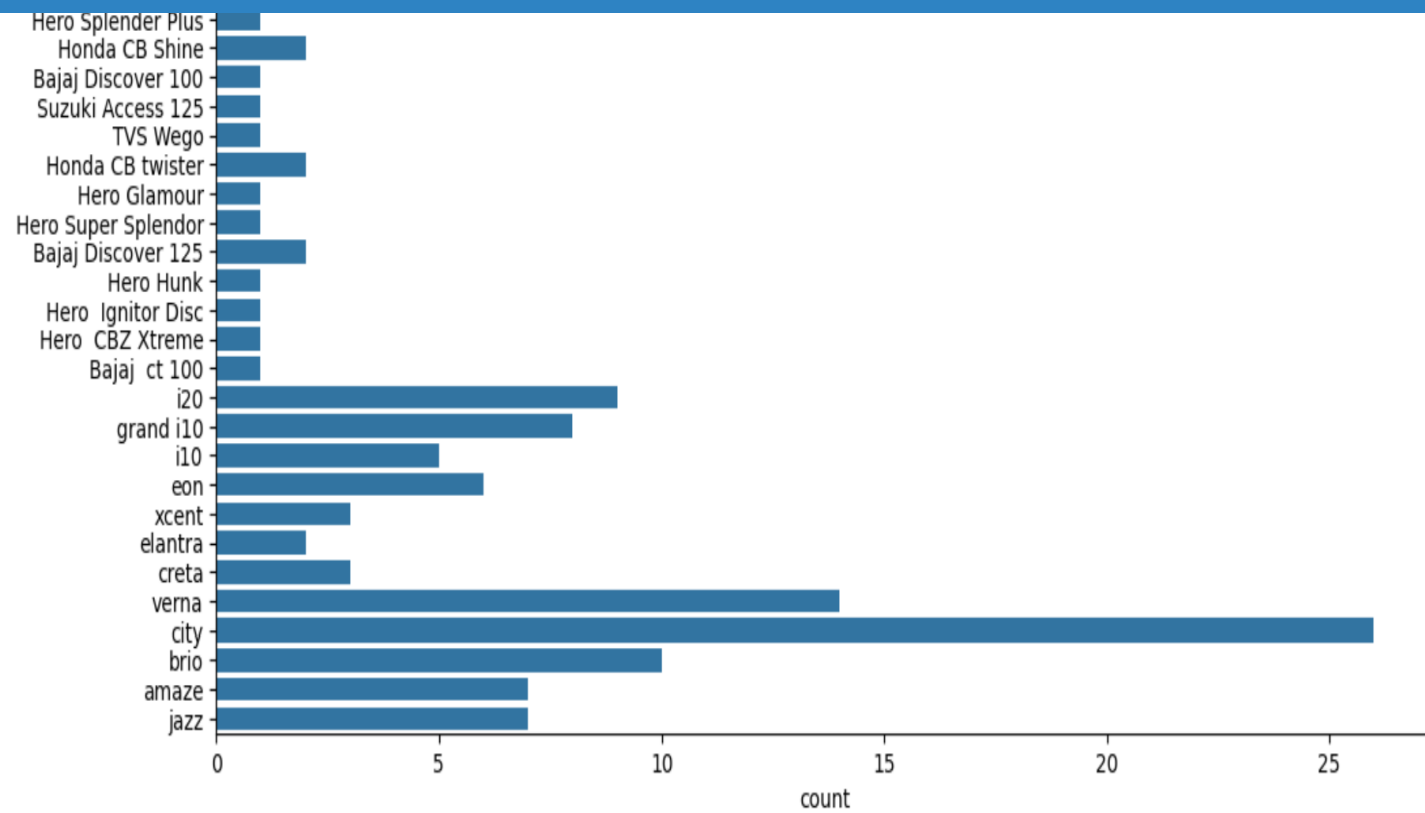
✓ 1s

```
fig_dims = (12, 20)
fig, ax = plt.subplots(figsize=fig_dims)
sns.countplot(y = data.Car_Name, ax=ax, data = data)
```

▼

<Axes: xlabel='count', ylabel='Car_Name'>





There are 98 different vehicles in the data. It contains biles and mopets too. Variable is categorical.

```
[12] print(data.Fuel_Type.value_counts())
```

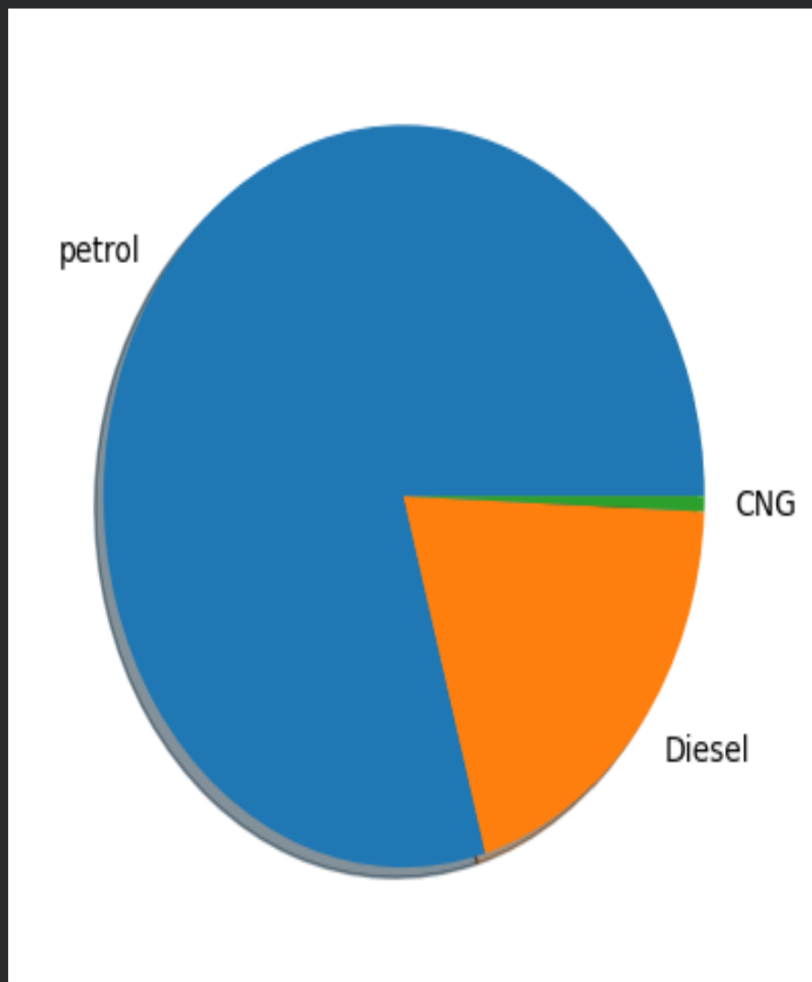
```
✓ 0s Fuel_Type  
Petrol    239  
Diesel    60  
CNG        2  
Name: count, dtype: int64
```



[14]

✓ 0s

```
plt.pie(data.Fuel_Type.value_counts(), labels=["petrol","Diesel","CNG"], shadow=True)  
plt.show()
```





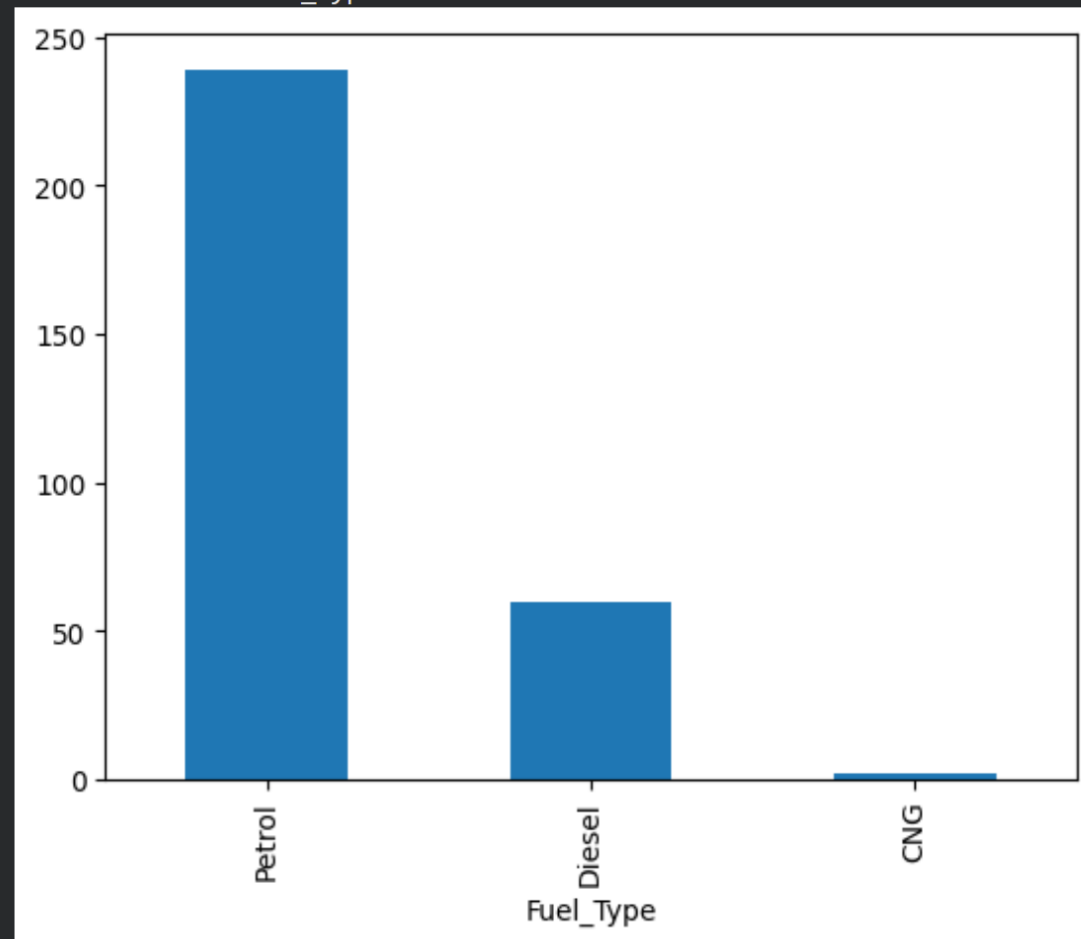
[15]
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```
data.Fuel_Type.value_counts().plot.bar()
```



... <Axes: xlabel='Fuel_Type'>



The data contains mostly petrol vehicles. This variable is skewed. Variable is categorical.

Thank you

Car Market Trends Analysis with Car Dekho Data

Presented by:
Yashveer Singh

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